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RTL to GDSII implementation of Advanced High-Performance Bus Lite

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Abstract— In this paper, it speaks about design of AHB-Lite and its implementation from RTL to GDSII. Cadence NC Launch tool is used for simulation. Input to NC Launch tool is Verilog file. Later gate level synthesis is carried out using cadence genus tool. Input to the cadence genus tool are i) verified Verilog file ii) .sdc file (synopsis design constraint file) iii) .lib file (library file).Output from cadence genus tool are gate level net list file and .sdc file (design constraint file).These both steps are of frontend part of design .And, in backend part of design cadence innovus tool is used. Reports for timing, area and power are calculated at all these steps of operations. At, last GDSII file is generated. All these process are carried out using 180nm technology cadence tool.

Advanced High-performance-Lite bus operations like read and write will be designed, simulated and synthesized .Later, the physical implementation of Advanced High-performance-Lite will be implemented on Cadence Innovus tool. Optimization is carried out all steps of design in terms of power, timing and area. After doing post CTS, there will be more optimization. One have obtained optimized results for slack 2.019ps, power 62.6nW and area 442747.1664μm^2.

Keywords—**RTL, CTS, AHB, GDSII, Innovus tool, NC Launch tool, Genus tool, sdc**

I. INTRODUCTION

AHB-Lite used, whenever the requirement of synthesizable designs of high performance. It is a best communication protocol to communicate between master and slave for higher rate date transfer capacity. This paper explains the design and implementation of an Advanced High-performance Bus -Lite bus. The main aim of the paper is to design Advanced High-performance Bus -Lite bus starting from register-transfer level to Graphic Database System Information Interchange (Physical Design) [1].

Figure 1 explains information regarding communication between single master and multiple slaves. Information for decoder will be given from master address lines and sent to Slaves. The multiplexers will get information from decoder to get select line information and send information from slaves to Master.

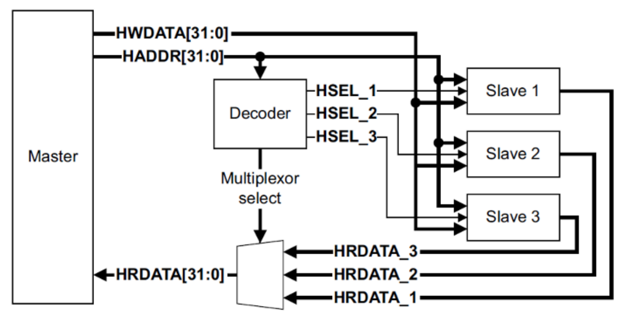

Figure 1: AHB-Lite Architecture

A. Master

In AHB-Lite, master gives information regarding read and write transaction. In paper, transaction related to busy transaction carried out.

B. Slave

An AHB-Lite slave will give response to transactions initiated by master. Information from decoder is used to communicate between master and slaves. Information from slave will be sent to master in the form of read transaction through multiplexers.

C. Decoder

AHB-Lite uses one centralized decoder to communicate between master and slaves. Decoder component will decode address of each transaction and also gives a select line information to multiplier to communicate to master.

D. Multiplexers

AHB-Lite uses one centralized multiplexers to communicate between slaves and master. Multiplexer's component will use information from decoder select line and sends information from slave to master.

[1] Proposed SoC interconnection protection through formal verification. [2] Defines information regarding design implementation of AHB bus protocol. Operations like read and write are carried out for AHB Lite bus protocol and verified using Xilinx tool. All operations are carried with respect to rising clock edge, so that this design can be fit into any of design flows.[4] Defines about low area and power SOC applications with energy efficient peripheral and system buses. A different low power application, which helps in communication between master and slave in low power application. Paper tells about 5X clock cycle reduction of transactions.

[8]Paper defines system level modelling of AHB-Lite subset of AHB, which are subsets of AMBA.

[9]Paper defines about verification of AHB Lite protocol. It mainly contains information regarding performance of AHB-Lite.[10]Paper defines about multiple arbitration technique with multichannel bus. Techniques used are like fixed priority and round robin.[11]Proposed paper defines about memory controller enhancement of AHB, SDRAM and DDR. It contains information regarding interaction between AHB-Lite bus and DDR SDRAM controller. [12]Proposed paper defines about conversion between WISHBONE Bus protocol and AHB bus protocol.

The remaining part of work is as follows: in Section II proposed work on AHB-Lite protocol and gave explanation regarding physical design. Section III describes how the proposed experimental results. In Section IV, results are presented. In the end, conclusions are provided in Section V.

II. WORK PROPOSED

Physical design is a process carried out in backend part of design. Physical design part includes portioning, floor planning and routing. In portioning, complex circuit will be reduced to simple part. In floor planning, for all block proper boundary will be assigned. Figure 2 shows a RTL to GDSII design flow. And in routing part, depending upon requirement local and global routings are carried out. At all stages performance will be analyzed and optimized. Proposed work contains both front end and back end part of design.

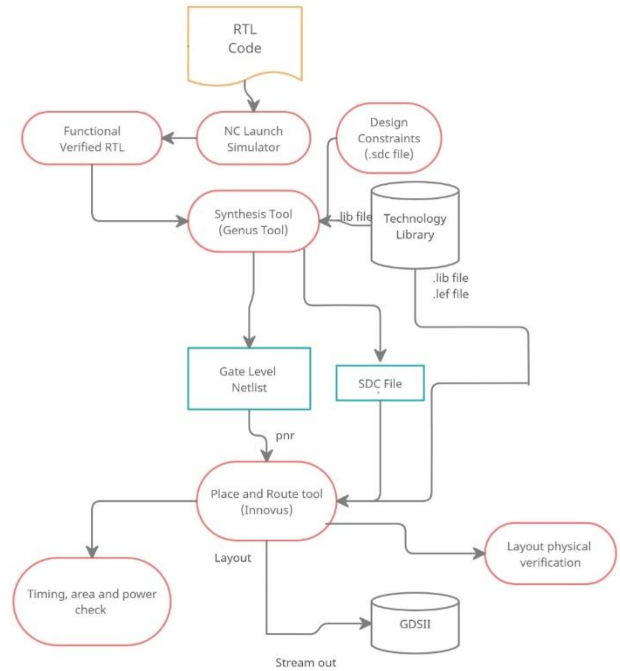

Figure 2: RTL to GDS II flow

Synthesis part:

Genus tool requires mainly 3 files

i) Verified Verilog file

ii).lib file (contains mainly timing information)

iii).sdc file (contains mainly information regarding clock period, clock latency)

Physical design part:

Innovus tool requires mainly 4 files

i) Gate level net list file obtained from genus tool.

ii).sdc file obtained from genus tool.

iii).lib file (library file like slow.lib and fast.lib)

iv).lef file (contains mainly information regarding metal layers of each simple logic blocks)

III. EXPERIMENTAL RESULTS

The outputs of a AHB-Lite with simple and busy transfer operation is carried out. Mainly focus on read and write operations. Figure 3 shows a Simulation results for read and write transaction with busy state for AHB-Lite operations.

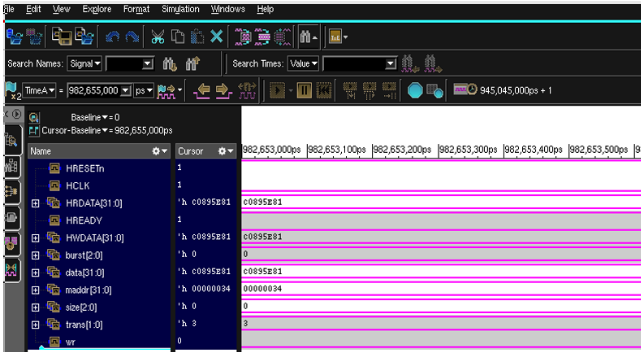

Figure 3: Simulation results of AHB-Lite

The Synthesis step includes genus tool. Genus tool will take Verilog file,.lib file and constraint file and provides output with 2 files as gate level net list verified Verilog file and .sdc (synopsis design constraint file).Later reports on area, power and timing are calculated. . Figure 4 shows a Gate-level net list of AHB-Lite.

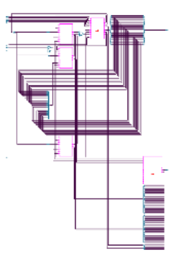

Figure 4. Gate-level net list of AHB-Lite

In back end part of process, first step is floor planning .where, height and width for floor planning will be provided. Later power strips are added at the boundaries of circuits. And at later part routing will be carried out. Reports for timing, area and power are obtained at each stages of operations. Figure 5 shows Physical Design of AHB-Lite.

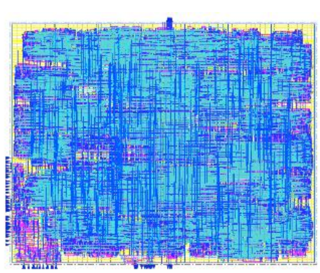

Figure 5: Physical Design of AHB-Lite

In back end part of process, Nano implantation of design will be carried out. Reports for timing, area and power are obtained at Nano implementation stages of operation. Figure 6 shows Nano implementation of AHB-Lite.

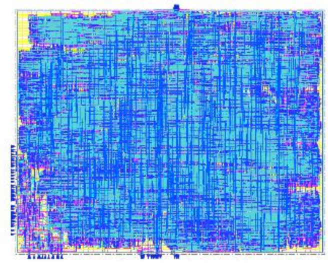

Figure 6: Nano implementation of AHB-Lite

In back end part of process, Optimized physical design implantation of design will be carried out. Reports for timing, area and power are obtained at Nano implementation stages of operation. Figure 7 shows Optimized Physical Design of AHB-Lite.

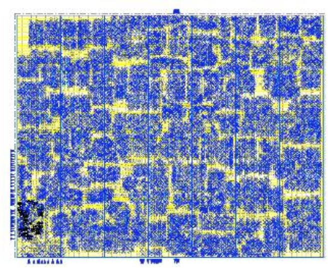

Figure 7: Optimized Physical Design of AHB-Lite

IV. ANALYSIS OF RESULT

Results mainly contains information regarding area, power and delay. In power, it contains information regarding leakage power, dynamic power and total power. In timing, it contains information regarding arrival time, required time and slack.

After doing synthesis, Table 1.defines information regarding slack, area consumed and total power consumption by design.

Table 1. Synthesis results

Timing(ps)	Area (um^2)	Total Power(nW)
Required=9784	826233	Internal Power=65.032
Arrival=10000		Switching Power=4.141
Slack= 1885		Leakage Power=4.931
		Total Power=74.104

After doing Pre-CTS, Table 2.defines information regarding slack, area consumed and total power consumption by design.

Table 2. Pre-CTS results

Timing(ps)	Area (um^2)	Total Power(nW)
Required=9.708	442747.1664	Internal Power=55.9259
Arrival=9.312		Switching Power=2.6038
Slack=0.396		Leakage Power=5.024
		Total Power=63.58

After doing post-CTS, Table 3.defines information regarding slack, area consumed and total power consumption by design.

Table 3. Post-CTS results

Timing(ps)	Area (um^2)	Total Power(nW)
Required=9.793	442747.1664	Internal Power=55.91
Arrival=7.774		Switching Power=1.661
Slack=2.019		Leakage Power=5.024
		Total Power=62.6

V. CONCLUSION

High bandwidth operations are provided by AMBA AHB-Lite. AHB-Lite supports optimal off chip and on chip connections with low power peripheral cells. AHB-Lite is designed and Implemented from RTL to GDSII, that includes both front end and back end designs. In front end, designing, simulation and gate level synthesis are carried out along with performance check. And, in back end, portioning, floor planning, routing and Nano routing are carried along with performance analysis. The Timing, Power and Area reports are generated in the synthesis flow and Physical Design flow. As there are trade off in circuits. In implementation, one have tried to optimize delay and power of design. Optimized results are as follows total power consumed by circuit: 62.6nW, total area consumed by circuit 442747.1664um^2 and optimized slack of circuit is 2.019ps. AMBA AHB supports high performance systems, high frequency clocks for transfer of data between master and slave.

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